It is a pleasure for me to be here, and I would like to express my appreciation to the organizers of the seminar for their invitation.

It is undoubtedly strange that an econometrician would propose to speak about survival analysis in large scale experiments on fruit flies. Many of my colleagues shake their heads as copies of papers from journals like *Experimental Gerontology* flow through our xerox machine. So I should begin with some explanation of this.

The leading demographer James Vaupel, in a recent letter to Science, made the following rather startling claim: Half the infants born this year in the US can expect to live to celebrate their 90th birthday. Related work suggests that half the girls born this year in North Dakota should live to see the year 2100. The dramatic demographic developments that underlie these forecasts have profound economic consequences....so, I tell my colleagues ... they should consider my work on medflies as just a small contribution to...saving Social Security. Some of you may have seen some reference to related research in Gina Kolata's piece two days ago in the Times, and in Ron Lee's piece in the new J. of Economic Perspectives.

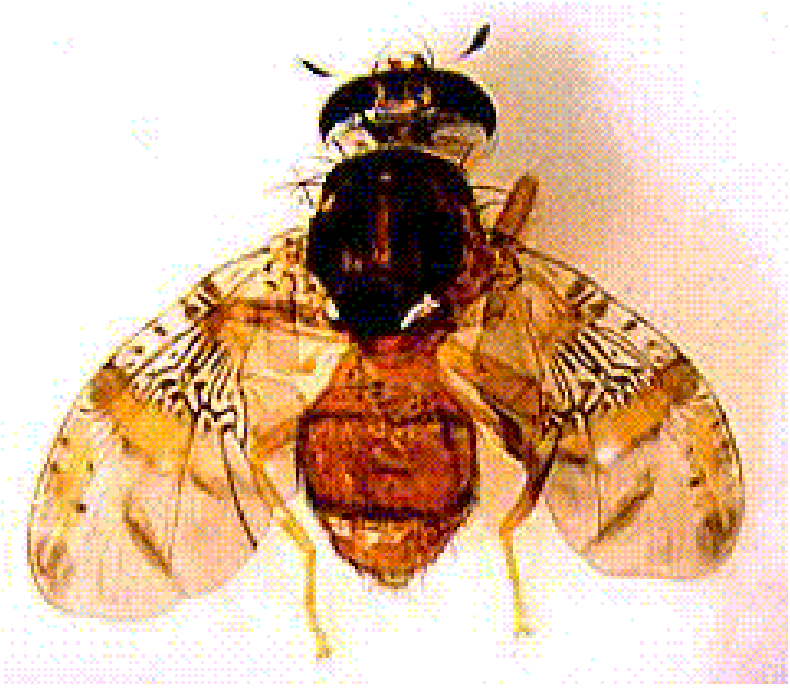
My real intent is more narrowly self-interested. I would like to offer a convincing case study on the use

of the quantile regression methods that I've been developing for survival analysis, thereby reinforcing arguments made by Chaudhuri, Doksum and Samarov in a recent Annals paper.

So here is a picture of our foe the mediterrean fruit fly, scourge of agricultural in Florida, California, and much of Latin America.

The talk will be based on joint work with Olga Geling, who is a student in the Statistics Dept. at Illinois. Olga has a strong interest in demography of aging, and was responsible for obtaining the data I will now briefly describe.

Reappraising Medfly Longevity: A Quantile Regression Survival Analysis



Roger Koenker

and

Olga Geling

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Carey Medfly Experiment

- 1,203,646 medflies survival times recorded in days
- Sex determined at time of death
- Pupae were sorted into one of five size classes
- 167 aluminum mesh cages of roughly 7200 flies each
- Adults were given a diet of sugar and water *ad libitum*

Life Cycle of Medfly

A third motivation for working on medflies is the almost irresistable attraction of a sample of 1.2 million uncensored survival times. The data we are working with is from the largest of several experiments on lower animals designed to investigate features of the survival distribution. The flies were raised in a large facility in Mexico. Pupae were sorted by size and assigned to one of 167 cages, monitored daily with deaths recorded in days and sex determined at time of death.

3 Important findings have emerged from the extensive work on this Three ieriment.

Medfly Findings

- Mortality rates actually *declined* at the oldest observed ages. contradicting the view that aging is an inevitable, monotone process of senescence.
- The right tail of the survival distribution was, at least by human standards, remarkably long.
- The experiment provided strong evidence for a crossover in gender specific mortality rates.

I will briefly consider each of these conclusions in turn.

Unconditional Hazard

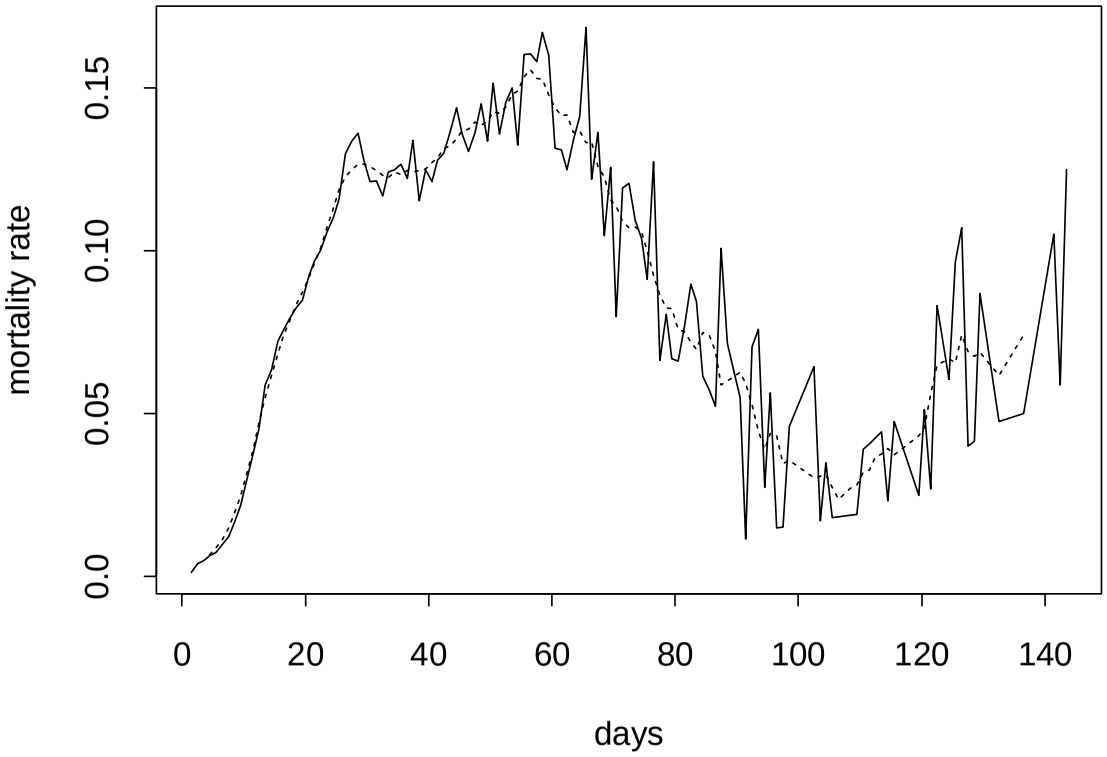

Figure 1: Raw mortality rate data for the full sample with a smoothed (7-day geometric moving average) estimate of the hazard function.

Medfly Survival Prospects

Lifespan (in days)	Percentage Surviving	Number Surviving
40	5	60,000
50	1	12,000
86	.01	120
146	.001	12

Human Survival Prospects*

Lifespan (in years)	Percentage Surviving	Number Surviving
50	98	591,000
75	69	413,000
85	33	200,000
95	5	30,000
105	.08	526
115	.0001	1

* Estimated Thatcher (1999) Model

Hazard by Gender

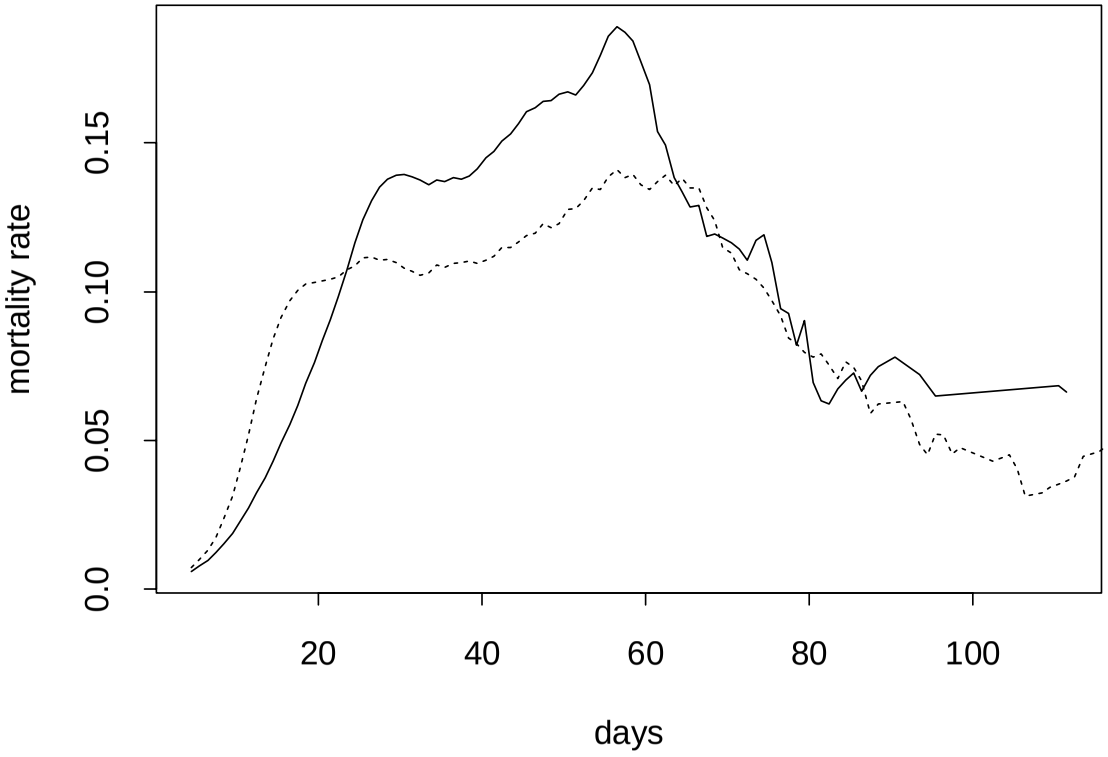

Figure 2: Smoothed mortality rates for males and females.

To give you some notion of the literature on this...here are a few references including the original Science paper, some controversy in the Science letters section, and some subsequent papers in other journals. One thing that is striking about much of this literature is the close parallels that are often drawn between the medfly work and issues in human demography.

Many of you will remember David Freedman's paper "From Mice to Man" about dose-response models in Statistical Science. We now need a new paper: "From Flies to Man" – Here is Jeff Goldblum in the starring role.

Some References

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From Fly to Man

So, with this introduction to medflies, let's talk a little about survival models. Kjell Doksum and Miriam Gasko, a few years ago wrote a nice ISR survey paper about the linkage between survival and binary response models.

Survival/Transformation Models

A wide variety of survival analysis models, Doksum and Gasko (1990), may be written as,

$$h(T_i) = x_i'\beta + u_i$$

where h is a monotone transformation,

T_i is an observed survival time,

x_i is a vector of covariates,

β is an unknown parameter vector

$\{u_i\}$ are iid with df F.

There are obviously many examples. I will briefly mention three.

Cox Model

For the proportional hazard model with

$$\log \lambda(t|x) = \log \lambda_0(t) - x'\beta$$

the conditional survival function in terms of the integrated baseline hazard $\Lambda_0(t) = \int_0^t \lambda_0(s)ds$ as,

$$\log(-\log(S(t|x))) = \log \Lambda_0(t) - x'\beta$$

so

$$\log \Lambda_0(T) = x'\beta + u$$

for u_i iid with df $F_0(u) = 1 - e^{-e^u}$.

Bennett Model

For the proportional odds model, where the conditional odds of death $\Gamma(t|x) = F(t|x)/(1 - F(t|x))$ are written as,

$$\log \Gamma(t|x) = \log \Gamma_0(t) - x'\beta,$$

we have

$$\log \Gamma_0(T) = x'\beta + u$$

for u iid logistic with $F_0(u) = (1 + e^{-u})^{-1}$.

Accelerated Failure Time Model

In the accelerated failure time model we have

$$\log(T_i) = x_i'\beta + u_i$$

so

$$\begin{aligned} P(T > t) &= P(e^u > te^{-x\beta}) \\ &= 1 - F_0(te^{-x\beta}) \end{aligned}$$

where $F_0(\cdot)$ denotes the df of e^u, and thus,

$$\lambda(t|x) = \lambda_0(te^{-x\beta})e^{-x\beta}$$

where $\lambda_0(\cdot)$ denotes the hazard function corresponding to F_0. In effect, the covariates act to rescale time in the baseline hazard.

The common feature in all these examples is the iid error assumption which effectively means that covariates exert a pure location shift effect on some transformation of the survival time. This seems unduly restrictive. One might even call it a "bug" not a "feature". A more general view of the treatment effect is nicely captured by an old, and seemingly rather neglected, notion introduced by Lehmann.

This is a natural way to begin to look at Quantile Regression, I think.

Quantile Treatment Effects

The simplest version of quantile regression is the two-sample treatment-control model,

Lehmann (1974) treatment response model:

> "Suppose the treatment adds the amount $\Delta(x)$ when the response of the untreated subject would be x. Then the distribution G of the treatment responses is that of the random variable $X + \Delta(X)$ where X is distributed according to F."

Doksum in a 74 Annals paper interprets the Lehmann QTE as the horizontal distance between the control and treatment marginal dfs, this means that it is really just what we would usually plot in the two-sample QQ-plot, or alternatively, it is just the difference between the two quantile functions.

Lehmann QTE as a QQ-Plot

Doksum (1974) defines $\Delta(x)$ as the "horizontal distance" between F and G at x, *i.e.*

$$F(x) = G(x + \Delta(x)).$$

Then $\Delta(x)$ is uniquely defined as

$$\Delta(x) = G^{-1}(F(x)) - x.$$

Changing variables so $\tau = F(x)$ we have the quantile treatment effect (QTE):

$$\delta(\tau) = \Delta(F^{-1}(\tau)) = G^{-1}(\tau) - F^{-1}(\tau).$$

This provides a nice connection to quantile regression since this difference is exactly what is estimated by the simplest conceivable two-sample quantile regression model.

QTE via Quantile Regression

The Lehmann QTE is naturally estimable by

$$\hat{\delta}(\tau) = \hat{G}_n^{-1}(\tau) - \hat{F}_m^{-1}(\tau)$$

where $\hat{G}_n$ and $\hat{F}_m$ denote the empirical distribution functions of the treatment and control observations, Consider the quantile regression model

$$Q_{Y_i}(\tau|D_i) = \alpha(\tau) + \delta(\tau)D_i$$

where D_i denotes the treatment indicator, and $Y_i = h(T_i)$, *e.g.* $Y_i = \log T_i$, which can be estimated by solving,

$$\min \sum_{i=1}^{n} \rho_\tau(y_i - \alpha - \delta D_i)$$

Why it is sometimes important to consider a more general notion of treatment effect than the mean-shift is nicely illustrated in Doksum's example.

Here we have two samples of guinea pigs, controls and treatments who received an injection of a virulent form of tubercle bacilli.

Surprisingly, the treatment has a beneficial effect on survival prospects in the lower tail, and a very detrimental effect in the upper tail.

This can't happen in conventional models.

Guinea Pigs

Doksum (1974) reconsiders a study by Bjerkedal (1960) on the effect of tubercle bacilli on guinea pigs.

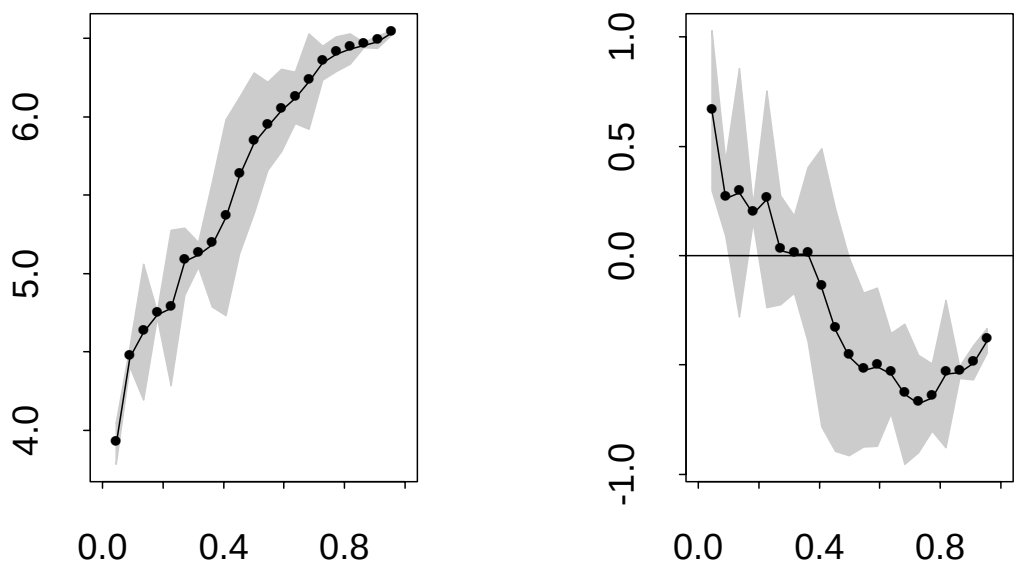

Figure 3: Quantile regression estimates: $\hat{\alpha}(\tau)$ in the right panel and $\hat{\delta}(\tau)$ in left.

Kaplan-Meier

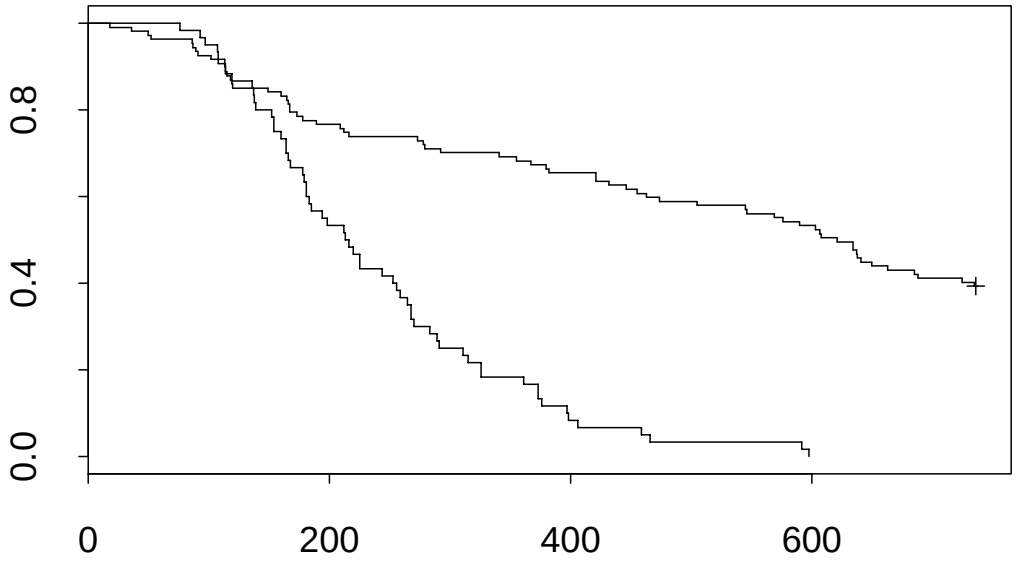

Figure 4: Kaplan-Meier estimates of the control and treatment guinea pig survival functions for the Bjerkedal data.

This idea extends nicely to general regression settings where the treatment variables are more complicated. Just as we can extend the unconditional mean estimate via minimizing sums of squares to estimation of conditional mean functions. We can extend estimation of unconditional quantiles to estimation of conditional quantile functions with an optimization formulation of the quantiles.

Computational Considerations

The problem of estimating $\beta(\tau)$ in our basic model

$$Q_{h(T)}(\tau|x) = x'\beta(\tau)$$

may be formulated as

$$\hat{\beta}(\tau) = \text{argmin}_{b \in \mathbb{R}^p} \sum_{i=1}^{n} \rho_\tau(h(T_i) - x_i' b)$$

where $\rho_\tau(u) = u(\tau - I(u < 0))$, (Koenker and Bassett (1978)). This is a linear program and is a natural extension of the one-sample definition of the τth sample quantile as any minimizer of the function,

$$R(a) = \sum_{i=1}^{n} \rho_\tau(y_i - a).$$

Monotone Equivariance

In the quantile regression model, for any monotone function, $h(\cdot)$, we have,

$$Q_{h(T)}(\tau|x) = h(Q_T(\tau|x)),$$

which follows immediately from observing that

$$P(T < t|x) = P(h(T) < h(t)|x).$$

This equivariance to monotone transformations of the conditional quantile functions is a crucial feature, allowing us to decouple the potentially conflicting objectives of transformations in conventional conditional mean regression.

The approach has a number of nice equivariance features including equivariance to monotone transformations of the response variable.

Inference in Quantile Regression I

The asymptotic behavior of the quantile regression process $\{\hat{\beta} : \tau \in (0,1)\}$ closely parallels the theory of ordinary sample quantiles in the one sample problem. In the classical linear model,

$$y_i = x_i\beta + u_i$$

with u_i iid from dfF, with density $f(u) > 0$ on its support $\{u|0 < F(u) < 1\}$, the joint distribution of $\sqrt{n}(\hat{\beta}_n(\tau_i) - \beta(\tau_i))_{i=1}^m$ is asymptotically normal with mean 0 and covariance matrix $\Omega \otimes D^{-1}$. Here $\beta(\tau) = \beta + F_u^{-1}(\tau)e_1, e_1 = (1, 0, \ldots, 0)'$, $x_{1i} \equiv 1, n^{-1}\sum x_i x_i' \to D$, a positive definite matrix, and

$$\Omega = ((\tau_i \wedge \tau_j - \tau_i\tau_j)/(f(F^{-1}(\tau_i))f(F^{-1}(\tau_j)))_{i,j=1}^m.$$

Inference can be based on asymptotics closely connected to familiar results for other M-estimators.

In non-iid error models we get a variant of the Huber Sandwich.

Inference in Quantile Regression II

When the response is conditionally independent over i, but not identically distributed, the asymptotic covariance matrix of $\zeta(\tau) = \sqrt{n}(\hat{\beta}(\tau) - \beta(\tau))$ is somewhat more complicated. Let $\xi_i(\tau) = x_i\beta(\tau)$, $f_i(\cdot)$ denote the corresponding conditional density, and define,

$$\begin{aligned} J_n(\tau_1, \tau_2) &= (\tau_1 \wedge \tau_2 - \tau_1\tau_2) n^{-1} \sum_{i=1}^{n} x_i x_i', \\ H_n(\tau) &= n^{-1} \sum x_i x_i' f_i(\xi_i(\tau)). \end{aligned}$$

Under mild regularity conditions on the $\{f_i\}$'s and $\{x_i\}$'s, we have joint asymptotic normality for $(\zeta(\tau_i), \ldots, \zeta(\tau_m))$ with covariance matrix

$$V_n = (H_n(\tau_i)^{-1} J_n(\tau_i, \tau_j) H_n(\tau_j)^{-1})_{i,j=1}^{m}.$$

Now returning to Medflies, I will briefly describe results for two basic models: In both cases the response variable is log of survival time.

Medfly Quantile Regression Models

- Model A:
  - SEX Gender
  - SIZE Pupal Size in mm
  - DENSITY Density of Cage
  - %MALE Initial Proportion of Males
- Model B:
  - BATCH + Pupal Cohort

More References

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In model A we have only 4 covariate effects: gender, size, cage density, and the effect of the initial proportion of males. The gender effect is quite interesting, as we have already seen. In this plot we have illustrate the 5 quantile regression coefficients as a function of tau in [0,1]. For gender you see that the effect is negative over most of the interval...females are coded 1, so this means that quantile of their survival distribution are below those of males until we get to the far right tail of the distribution – after tau =.95 there is a positive effect indicating that females have better survival prospects than males for the upper 5 percent of the population.

Size doesn't seem to matter very much, perhaps bigger flies are more robust early in life, later it seems larger flies are at a disadvantage.

Density has been a rather controversial effect the literature on the experiments on medflies. There is a strong presumption that it is a negative influence on longevity, but surprisingly it exerts a strong postive effect here. This is an effect that is difficult to analyse with the traditional life table methods employed in the original literature on the experiment.

Similarly, we thought that we should look at the effect of the initial proportional of males in the cage, and this too had a surprisingly strong positive effect.

In tests, this effect was essentially the same on both males and females. A rather puzzling result.

Model A

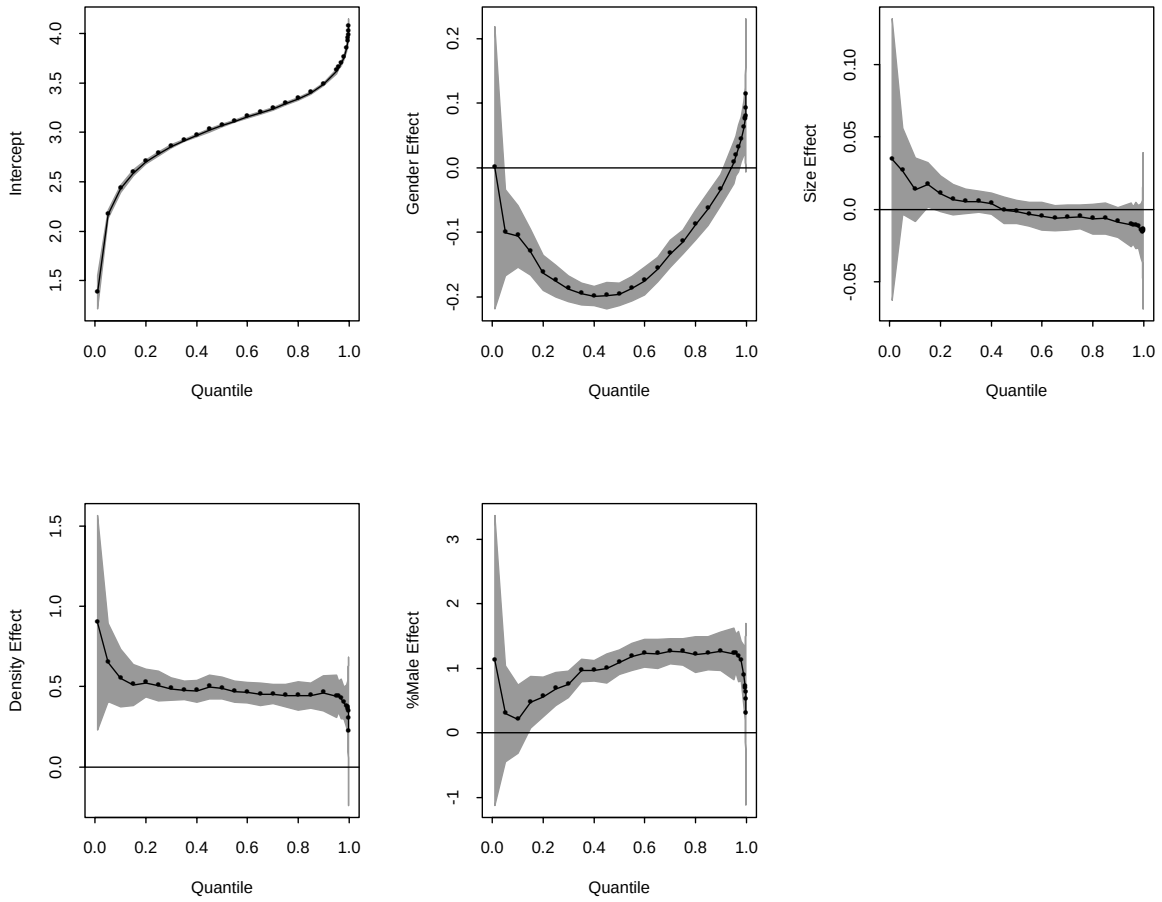

Figure 5: Quantile regression results for Model A. The lightly shaded region is a 90 percent pointwise confidence band for the corresponding coefficient.

When we introduce the cohort or batch effect we find a.) it is highly significant, and b.) it introduces some interesting modifications of the earlier results. This is easiest to see in the plot of the 5 main effects excluding batch...Now the gender effect looks very much like it did before, but the size effect is stronger, with bigger flies doing significantly better early on, but disadvantaged later in life.

The density effect is now only significant early on, where it is still beneficial, and per cent males is still a positive influence on survival, but the effect is weaker, and less precisely estimated.

Model B

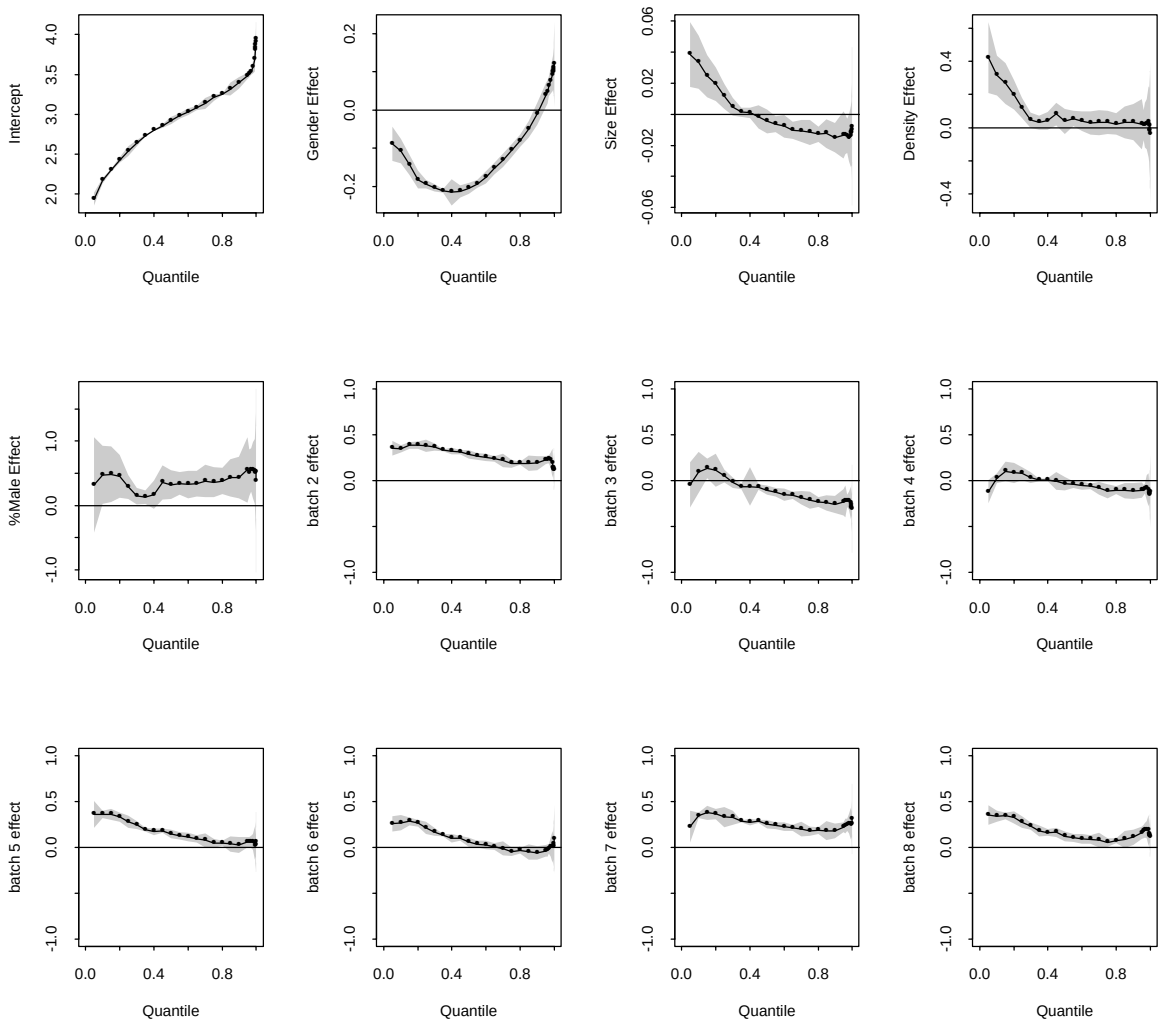

Figure 6: Quantile regression results for Model B.

Model B

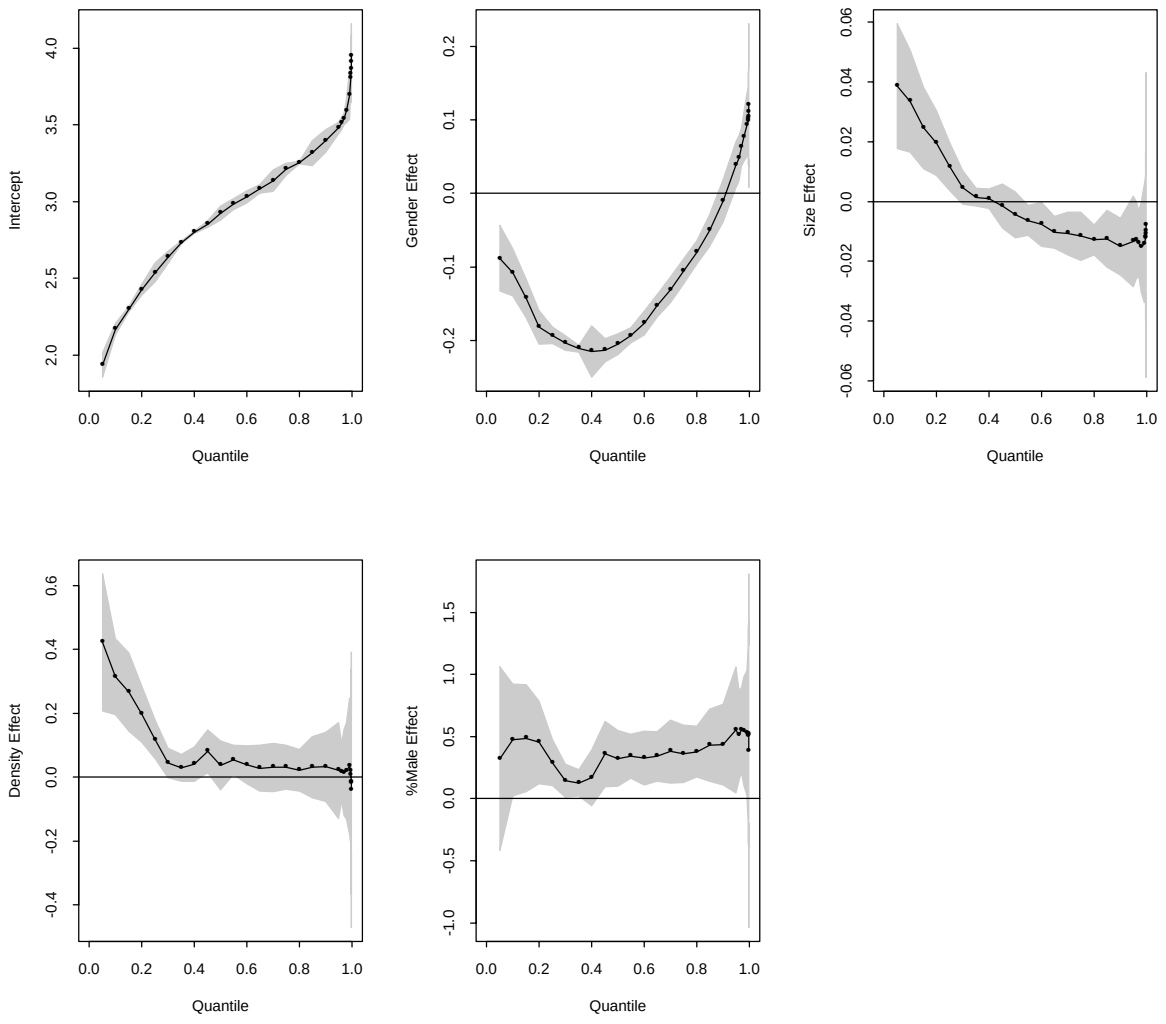

Figure 7: Quantile regression results for Model B.

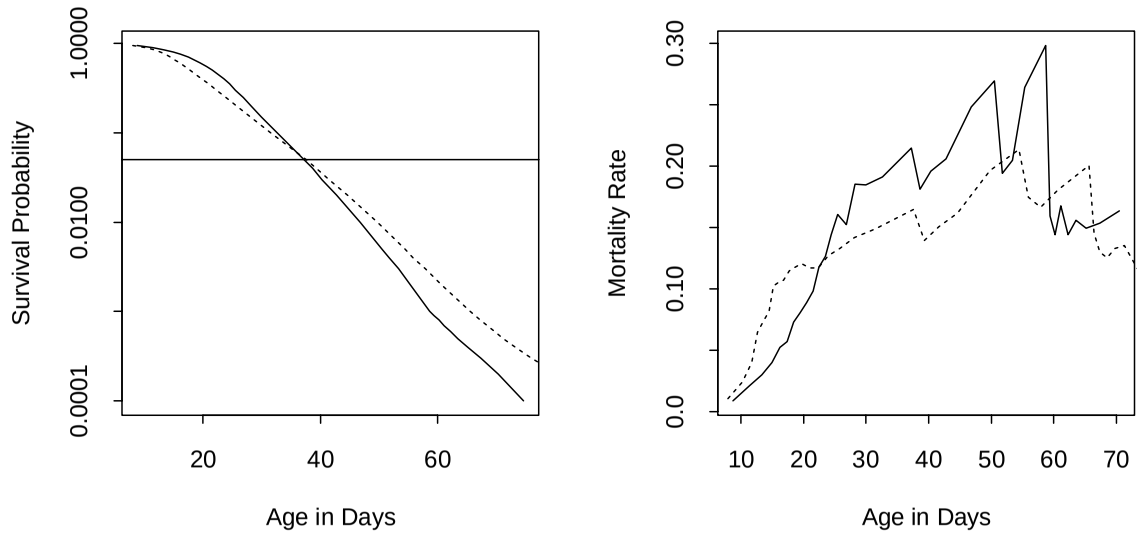

Figure 8: Estimated survival and hazard functions for Model A. Survival and hazard functions are illustrated males (solid line) and females (dotted line) with the other covariates evaluated at the experimental sample means. Horizontal line indicates $\hat{S}(t) = .05$.

It is possible starting from the quantile regression estimates to compute corresponding estimates of variance other functions of interest. Here we consider survival curves based on model A and the corresponding hazard functions for males and females. Note the crossing at about38 days corresponds to the .95 quantile of the survival distribution. Again, we see that estimated hazards decline after about 60 days.

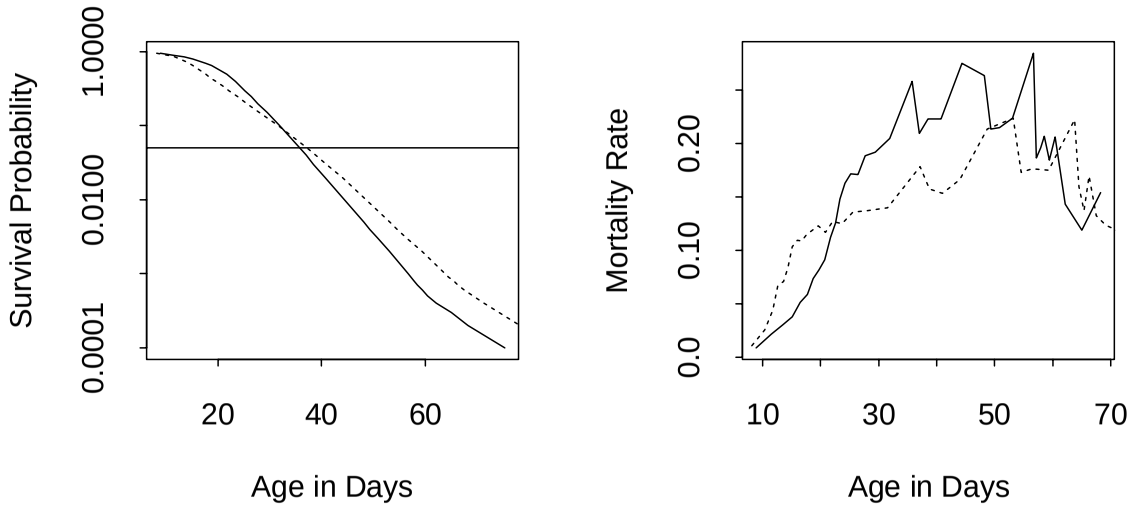

Figure 9: Estimated survival and hazard functions for Model B. Survival and hazard functions are illustrated males (solid line) and females (dotted line) with the other covariates evaluated at the experimental sample means. Horizontal line indicates $\hat{S}(t) = .05$.

To compare with more conventional methods we have estimated a Cox model corresponding to Model A.

A Proportional Hazard Model

To contrast the quantile regression results with the Cox model we estimated the following proportional hazard (PH) model corresponding to model A.

$$\begin{aligned}\log \lambda(t|x) &= \log \lambda_0(t) + \underset{(0.0018)}{0.2165}\mathsf{SEX} + \underset{(0.00084)}{0.0124}\,\mathsf{SIZE} \\ &- \underset{(0.00072)}{1.021}\,\mathsf{DENSITY} \underset{(0.0203)}{-2.625}\,\%\mathsf{MALE}.\end{aligned}$$

All four covariate effects constitute a rescaling of the baseline hazard function, $\lambda_0(t)$.

But in the Cox model all the covariates have the same shape to the effect illustrated in the first panel and impose the condition that the effects can't cross zero, so the hazard functions are scale shifts of one another. In this case this means that there can be no cross-over in the gender effect.

Cox Treatment Effects

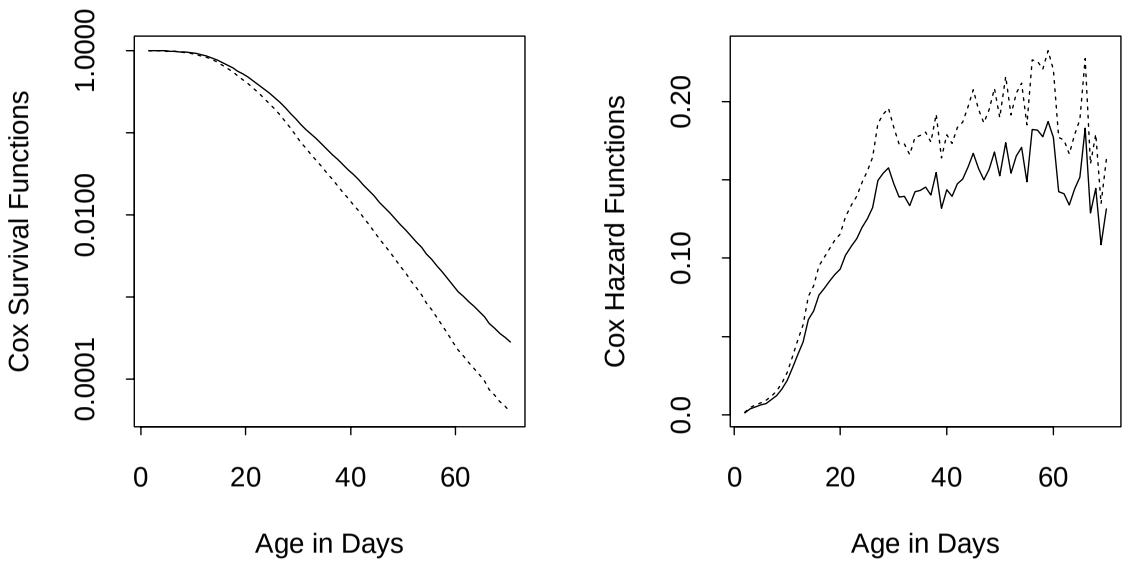

Figure 10: Estimated Cox quantile treatment effect implicit in the estimated Cox model. Estimated hazard functions for males and females based on the Cox proportional hazard model.

5 Medfly Life Lessons

Males are tough ...	but only until 40.
Bigger is better ...	but only before 18.
Small is beautiful...	after 25.
Crowds are good ...	especially of guys.
Life gets safer ...	but only after 60.

Quantile Regression Survival Models

- More flexible semiparametric survival models.
- Scalar treatment effects $\rightarrow$ functions.
- Natural interpretability of quantiles.
- Attractive equivariance and computation.
- Open problems: Inference, censoring, ...